

33)

$$e^x - 2 = 0$$

$$x_0 = 1$$

$$f'(x) = e^x$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = x_n - \left(\frac{e^{x_n} - 2}{e^{x_n}} \right)$$

$$x_{n+1} = x_n - 1 + 2 \cdot e^{-x_n}$$

34)

$$2x + 3y = 6$$

$$4x + 6y = 3k$$

$$\left[\begin{array}{cc|c} 2 & 3 & 6 \\ 4 & 6 & 3k \end{array} \right]$$

$$R_2 - 2R_1$$

$$\left[\begin{array}{cc|c} 2 & 3 & 6 \\ 0 & 0 & 3k - 12 \end{array} \right]$$

$$\rightarrow \text{Rank} = 1$$

for system to have non-zero sol

$$3k - 12 = 0$$

$$k = 4$$